

Chapter 1

Probability Theory

1.1 Expectation

Non-Equal Likelihood

Let X be a random variable with a finite number of outcomes x_i for $i = 1, \dots, n$ with probability p_i . The **expectation** of X is defined as

$$\mathbb{E}[X] \triangleq \sum_{i=1}^n p_i x_i. \quad (1.1)$$

Because the sum of all probabilities is identically equal to unity,

$$\sum_{i=1}^n p_i = 1.0, \quad (1.2)$$

the expected value is considered the **weighted average**, where each p_i is the **weight**.

Remark 1.1.1. The weighted average is also known as the **first moment**, which is analogous to the **center of mass** of a rigid body.

Equal Likelihood

For the special case where each probability is equally likely,

$$p_1 = p_2 = \dots = p_n = \frac{1}{n}, \quad (1.3)$$

the expectation of $\mathbb{E}[X]$ has the specialized form μ ,

$$\mu \triangleq \mathbb{E}[X] \stackrel{(1.3)}{=} \frac{1}{n} \sum_{i=1}^n x_i. \quad (1.4)$$

and is referred to as the **simple average**.

Remark 1.1.2. The expectation $\mathbb{E}[X]$, also known as the weighted average of X , should not be confused with μ , the simple average of X . The latter is a special case of the former, as just demonstrated in the above sections.

1.2 Variance

1.2.1 Non-Equal Likelihood

The **Variance** of X is defined as

$$\text{VAR}(X) \triangleq \mathbb{E}[(X - \mathbb{E}[X])^2], \quad (1.5)$$

$$\begin{aligned} & \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])], \\ & \mathbb{E}[X^2 - \mathbb{E}[X]X + \mathbb{E}[X]\mathbb{E}[X]], \\ & \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[X], \\ & \mathbb{E}[X^2] - \mathbb{E}[X]. \end{aligned} \quad (1.6)$$

Equal Likelihood

For the special case where each probability is equally likely, (1.6) takes the form

$$\sigma^2 \triangleq \mathbb{E}[X] \stackrel{(1.6)}{=} \sum_{i=1}^n p_i x_i^2 - \mu^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \mu^2, \quad (1.7)$$

and is referred to as the **simple variance**.

Remark 1.2.3. The form of (1.7) is often stated in slightly different form,

$$\frac{1}{n} \sum_{i=1}^n x_i^2 - \mu^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - 2\mu^2 + \mu^2, \quad (1.8)$$

$$\stackrel{(1.4)}{=} \frac{1}{n} \sum_{i=1}^n x_i^2 - \frac{2\mu}{n} \sum_{i=1}^n x_i + \frac{n\mu^2}{n}, \quad (1.9)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i^2 - 2\mu x_i + \mu^2), \quad (1.10)$$

$$= \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2. \quad (1.11)$$

This form, while perhaps easier to conceptualize, is relatively less efficient to implement since the latter requires n subtractions whereas the former requires only a single subtraction.

1.3 Covariance

Non-Equal Likelihood

For two jointly distributed real-valued random variables X and Y , with **expected values**¹ $\mathbb{E}[X]$ and $\mathbb{E}[Y]$, respectively, the **covariance** is defined as

$$\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])], \quad (1.12)$$

$$\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] - \mathbb{E}[X] \mathbb{E}[Y] + \mathbb{E}[X] \mathbb{E}[Y], \quad (1.13)$$

$$\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]. \quad (1.14)$$

Let the random variable pair (X, Y) take on values (x_i, y_i) for $i = 1, \dots, n$ with probability p_i . The covariance is defined as

$$\text{Cov}(X, Y) \triangleq \sum_{i=1}^n p_i (x_i - \mathbb{E}[X]) (y_i - \mathbb{E}[Y]). \quad (1.15)$$

Non-Equal Likelihood

If the probability of each outcome is equal, the $p_i = 1/n$, and (1.15) becomes

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (x_i - \mathbb{E}[X]) (y_i - \mathbb{E}[Y]). \quad (1.16)$$

1.3.1 Pearson Correlation Coefficient

The **Pearson correlation coefficient** (PCC)² is a measure of the linear correlation between two variables X and Y .

When a pair of random variables (X, Y) are applied to a population, the PCC is called the **population correlation coefficient** and denoted as $\rho_{X,Y}$ and is defined as

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad (1.17)$$

¹Also known as **mean values**.

²Also known as the Pearson's r coefficient, Pearson product-moment correlation coefficient (PPMCC), and the bivariate correlation.

where

- Cov is the covariance of X and Y ,
- σ_X is the standard deviation of X , and
- σ_Y is the standard deviation of Y .

The PCC is bounded as

$$-1 \leq \rho_{X,Y} \leq 1. \tag{1.18}$$

- A value of -1 indicates complete negative linear correlation.
- A value of 0 indicates the absence of linear correlation.
- A value of +1 indicates complete positive linear correlation.

Remark 1.3.4. The covariance of a variable with itself is simply the variance of the variable. So, $\text{Cov}(X, X) = \text{Var}(X)$.

Remark 1.3.5. When $\mathbb{E}[XY] \approx \mathbb{E}[X] \mathbb{E}[Y]$, then the $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]$ is prone to so-called “catastrophic cancellation” with floating point calculations. A numerically stable algorithm should be used.

Chapter 2

Modern Portfolio Theory

2.1 Cross-Correlation

In signal processing, **cross-correlation**¹ is a measure of similarity of two signals.

In probability theory, cross-correlation refers to the correlations between two random variables X and Y .

2.2 Portfolio and Assets

Let a portfolio P be a collection of n individual assets with present values $a_1, a_2, \dots, a_n$. Assets have units of U.S. dollars (USD).²

The present valuation of the portfolio, $PV(P)$, is given as

$$PV(P) = \sum_{i=1}^n a_i. \quad (2.1)$$

2.3 Portfolio Weighting

The relative weight w_i of each asset a_i is simply the fraction of the asset's present value to the portfolio's present value, thus

$$w_i = \frac{a_i}{PV} = \frac{a_i}{\sum_{i=1}^n a_i}. \quad (2.2)$$

¹Cross-correlation is also called the **sliding dot product** or **sliding inner product**

²When the asset is stock, bond, or other instrument that consists of fractional ownership (shares) denominated by price per share, then the present value of the asset is simply the number of shares held (holdings) in the portfolio multiplied by the price per share of that holding.

The sum of all weights, must equal one (1).

$$\sum_{i=1}^n w_i = \sum_{i=1}^n \frac{a_i}{PV} = \frac{1}{PV} \sum_{i=1}^n a_i = \frac{PV}{PV} = 1. \quad (2.3)$$

2.4 Portfolio Return

Let $\mathbb{E}(R_P)$ and $\mathbb{E}(R_i)$ be the expected periodic return R_P of the portfolio P and expected periodic return R_i on an asset a_i , respectively.³ The expected portfolio return is then simply the weighted sum of each asset's return,

$$\mathbb{E}(R_P) = \sum_{i=1}^n w_i \mathbb{E}(R_i). \quad (2.4)$$

Let the mean periodic return $\bar{R}(a_i)$ of asset a_i is given by observing the return at K times during the periodic return interval,

$$\bar{R}(a_i) = \frac{1}{K} \sum_{k=1}^K R_k(a_i). \quad (2.5)$$

For clarity, the notation $f(a_i)$ to denote *on a per-asset basis* is assumed, as written hereinafter simply as f . Then the mean periodic return is given as

$$\bar{R} = \frac{1}{K} \sum_{k=1}^K R_k. \quad (2.6)$$

Let $\text{VAR}(R) = s^2$ be the (sample) variance of periodic returns on an asset a_i , given by

$$\text{VAR}(R) = s^2 = \frac{1}{K-1} \sum_{k=1}^K (R_k - \bar{R})^2. \quad (2.7)$$

The (sample) standard deviation s is the positive square root of the (sample) variance s^2 ,

$$s = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (R_k - \bar{R})^2}. \quad (2.8)$$

³The period is typically stated annually, quarterly, or yearly.